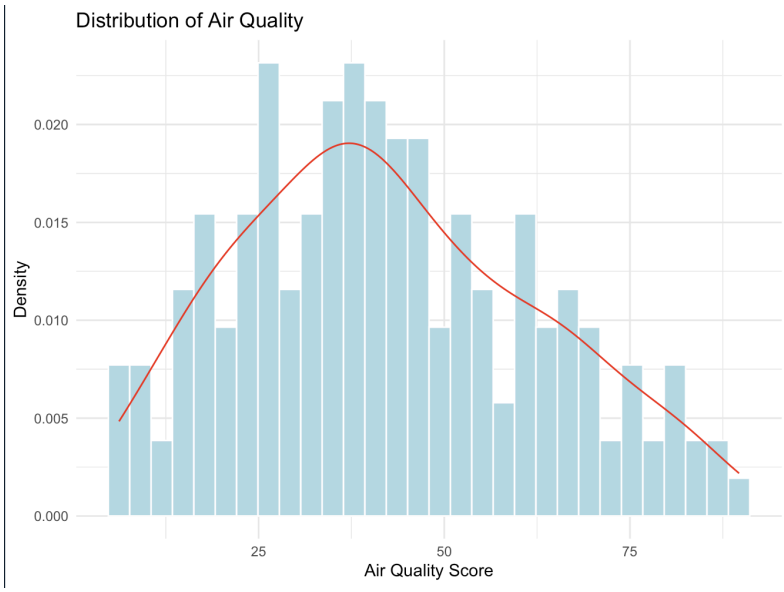
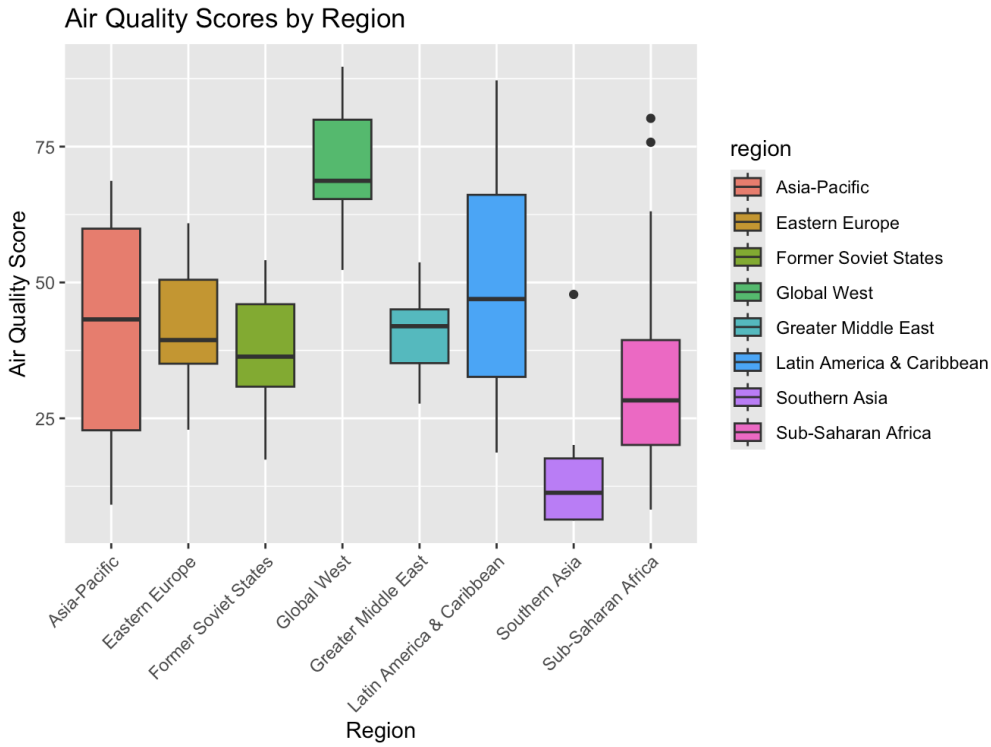
Variable Distributions

1.1



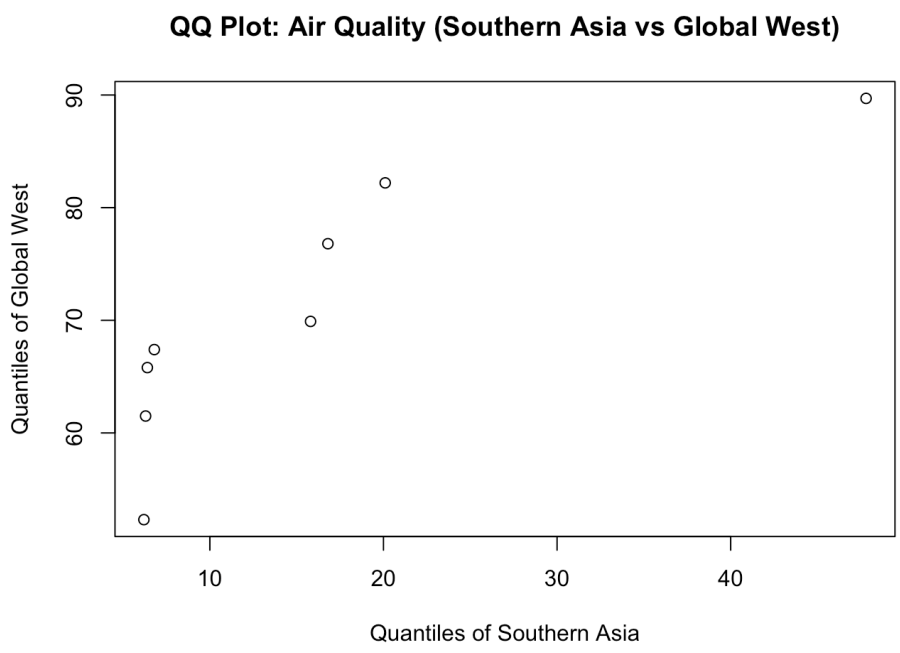
1.2



2.1

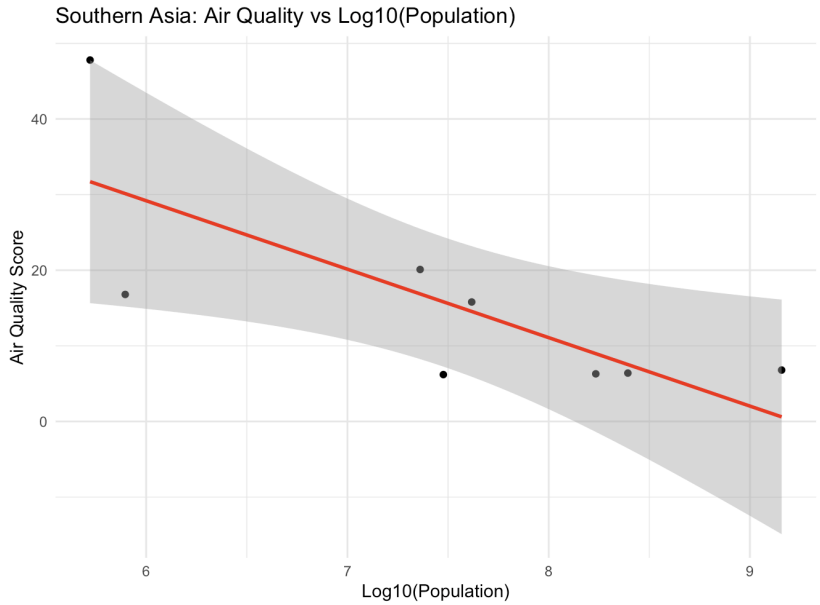
Air Quality Comparison: Southern Asian vs Global West
Count
0
1
2
3
4
5
25
50
75
Air Quality Score
region
Global West
Southern Asia

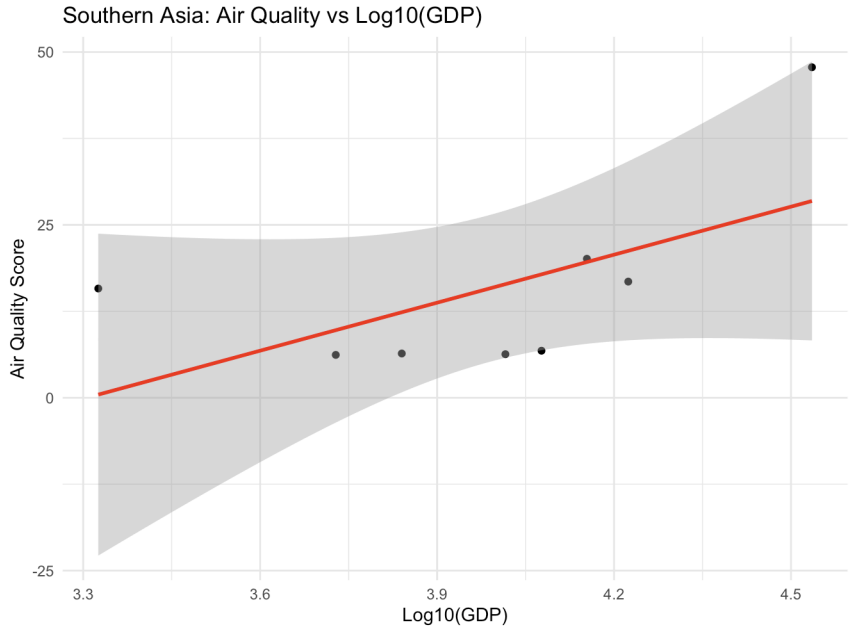
2.2

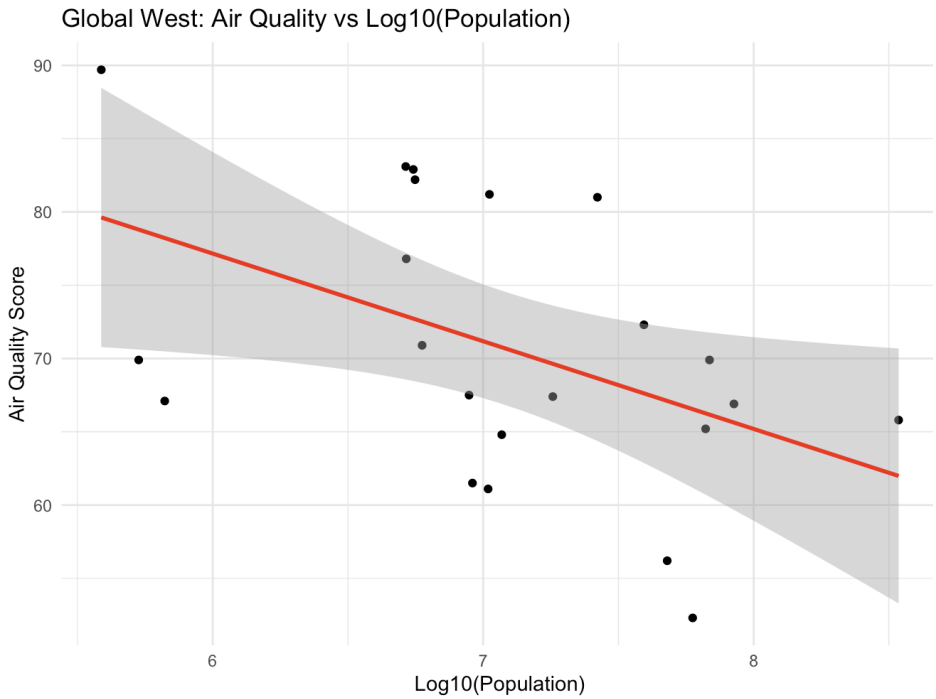

QQ Plot: Air Quality (Southern Asia vs Global West)
Quantiles of Global West
60
70
80
90
10
20
30
40
Quantiles of Southern Asia

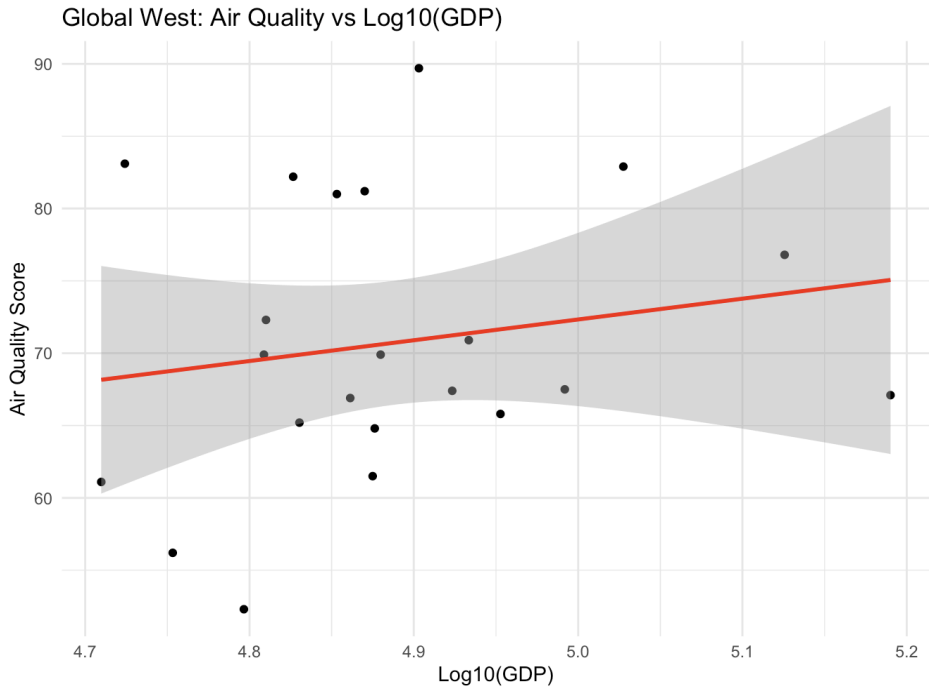
Linear Models

3.1






Global West: Air Quality vs Log10(Population)
Air Quality Score
90
80
70
60
6
7
8
Log10(Population)


Global West: Air Quality vs Log10(GDP)
Air Quality Score
90
80
70
60
4.7
4.8
4.9
5.0
5.1
5.2
Log10(GDP)

3.2

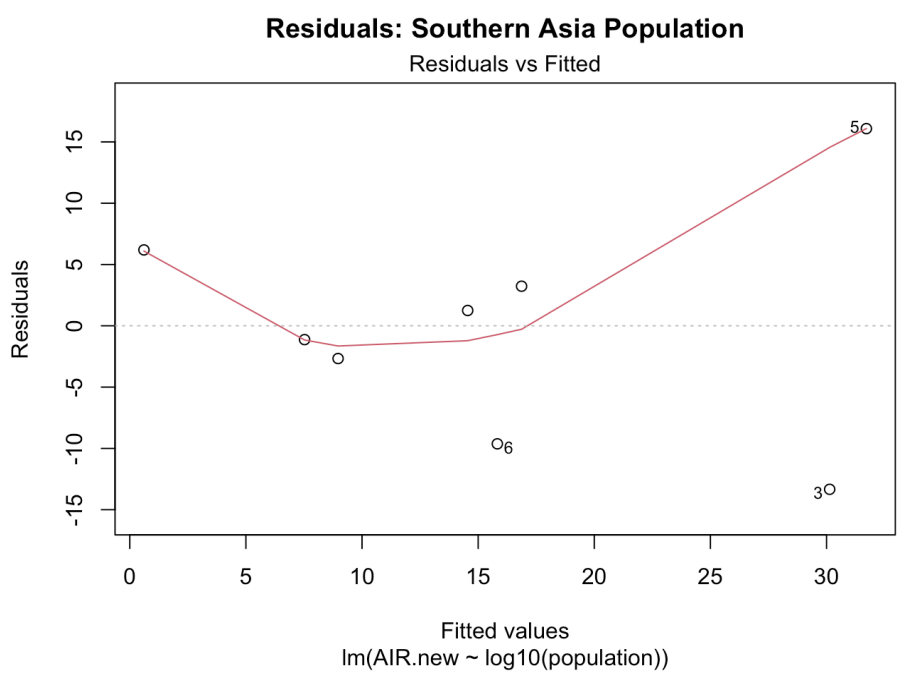
```
> # Fit Model 1: AIR Quality predicted by Population
> lm1.SA <- lm(AIR.new ~ log10(population), data = subset1)
> summary(lm1.SA)

Call:
lm(formula = AIR.new ~ log10(population), data = subset1)

Residuals:
     Min       1Q   Median       3Q      Max 
-13.3348  -4.4080   0.0621   3.9700  16.0845 

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)  
(Intercept)         83.496     23.850   3.501   0.0128 *
log10(population)   -9.051      3.153  -2.871   0.0284 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 9.899 on 6 degrees of freedom
Multiple R-squared:  0.5787,	Adjusted R-squared:  0.5084 
F-statistic:  8.24 on 1 and 6 DF,  p-value: 0.02841
```



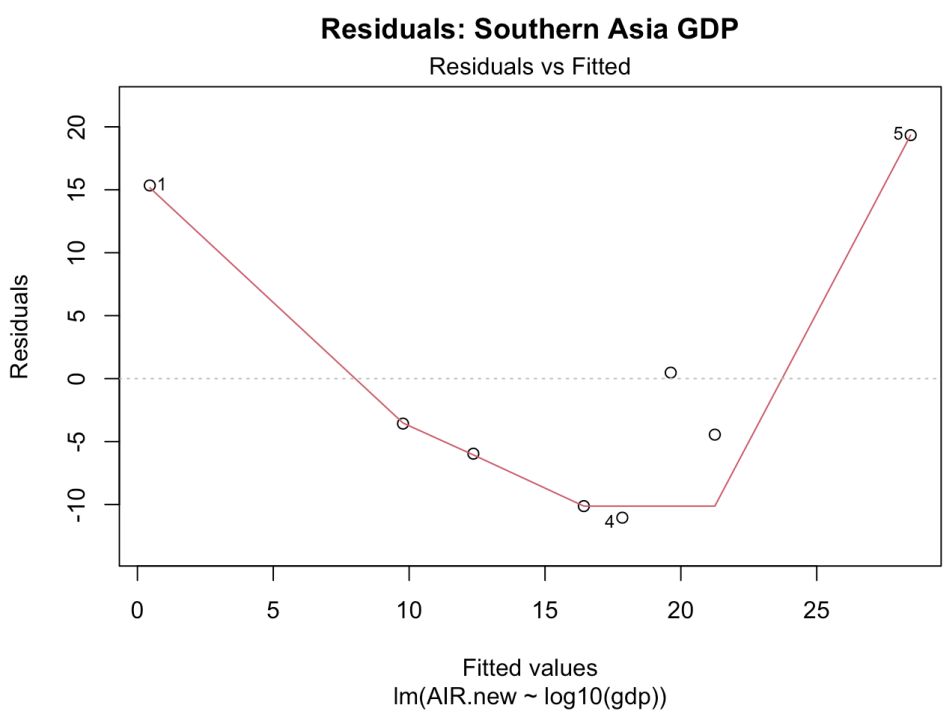
```
> # Fit Model 2: AIR Quality predicted by GDP
> lm2.SA <- lm(AIR.new ~ log10(gdp), data = subset1)
> summary(lm2.SA)

Call:
lm(formula = AIR.new ~ log10(gdp), data = subset1)

Residuals:
    Min      1Q  Median      3Q     Max 
-11.046  -7.005  -4.012   4.191  19.343 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   -76.50      51.17  -1.495    0.186
log10(gdp)     23.14      12.79   1.810    0.120

Residual standard error: 12.26 on 6 degrees of freedom
Multiple R-squared:  0.3531,	Adjusted R-squared:  0.2453 
F-statistic: 3.275 on 1 and 6 DF,  p-value: 0.1203
```



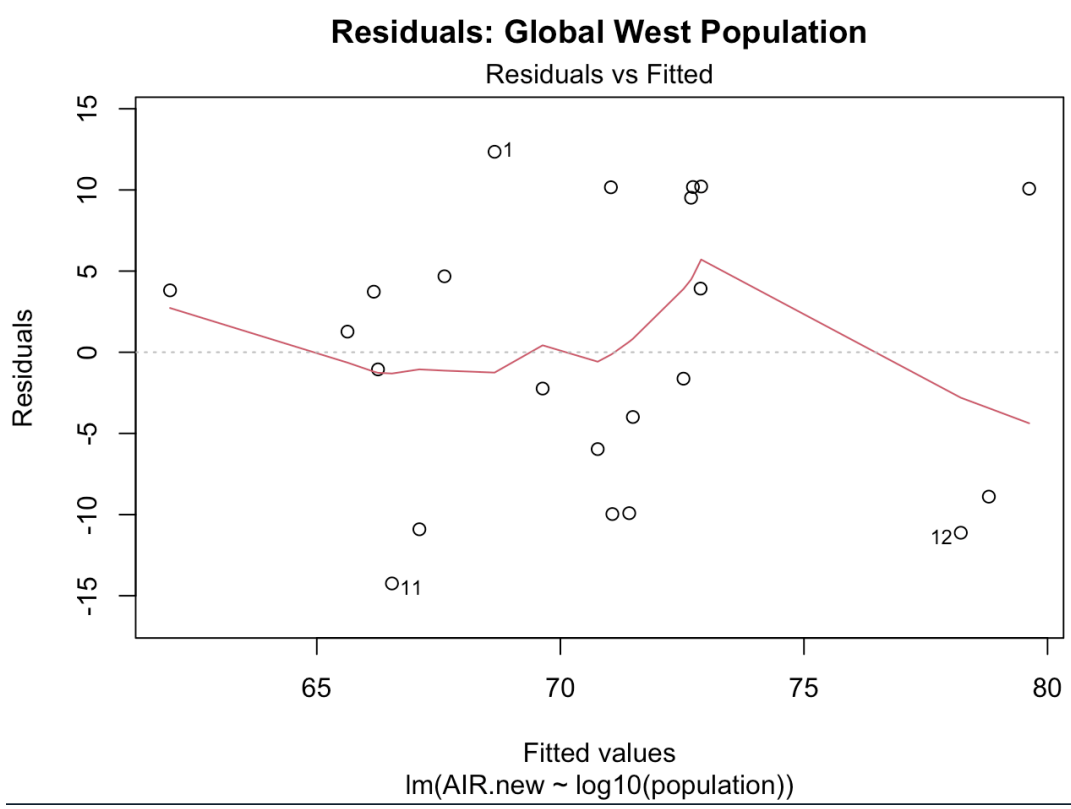
```
> # Fit Model 1: AIR Quality predicted by Population
> lm1.GW <- lm(AIR.new ~ log10(population), data = subset2)
> summary(lm1.GW)

Call:
lm(formula = AIR.new ~ log10(population), data = subset2)

Residuals:
     Min       1Q   Median       3Q      Max 
-14.2424  -8.1628   0.1074   8.3079  12.3510 

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)    
(Intercept)        113.060     18.219   6.205 4.63e-06 ***
log10(population)   -5.983      2.561  -2.336     0.03 *  
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 8.676 on 20 degrees of freedom
Multiple R-squared:  0.2144,	Adjusted R-squared:  0.1751 
F-statistic: 5.458 on 1 and 20 DF,  p-value: 0.03
```



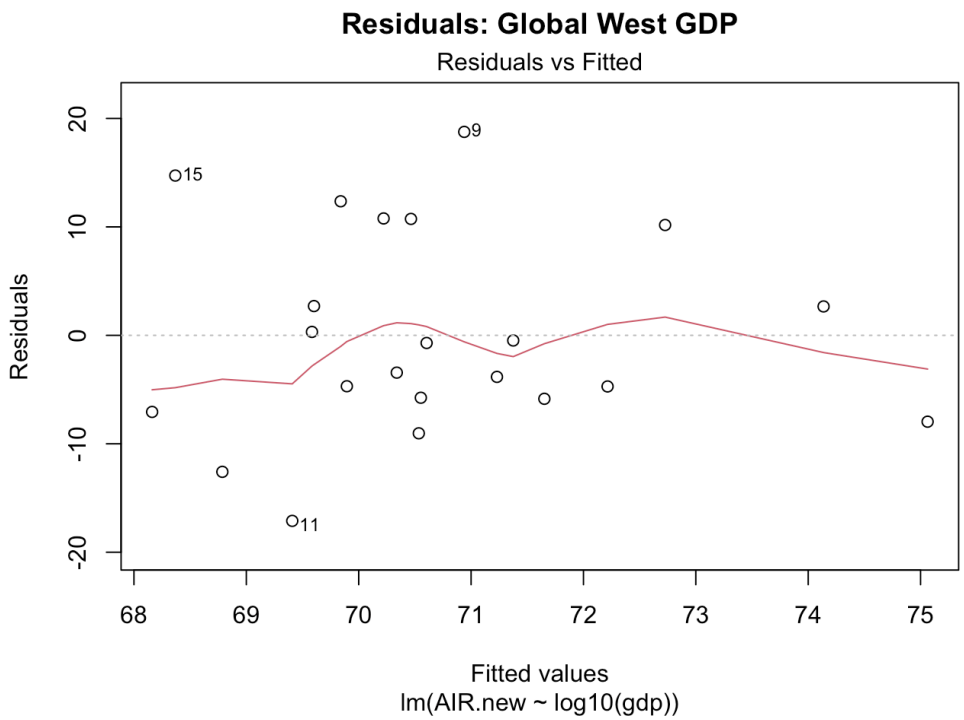
```
> # Fit Model 2: AIR Quality predicted by GDP
> lm2.GW <- lm(AIR.new ~ log10(gdp), data = subset2)
> summary(lm2.GW)

Call:
lm(formula = AIR.new ~ log10(gdp), data = subset2)

Residuals:
    Min      1Q  Median      3Q     Max 
-17.109  -5.827  -2.071   8.304  18.761 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.4935    87.0182   0.006    0.996
log10(gdp)   14.3676    17.7996   0.807    0.429

Residual standard error: 9.633 on 20 degrees of freedom
Multiple R-squared:  0.03155,	Adjusted R-squared:  -0.01687 
F-statistic: 0.6515 on 1 and 20 DF,  p-value: 0.4291
```



3.3

For both regions, the model that uses log10(population) as the predictor is the better one because the p value was only statistically significant using that predictor ($p < 0.05$ for population but $>> 0.05$ for GDP). Additionally, for both regions as well, the model that uses log10(population) as the predictor has a higher Multiple R-squared value than the GDP models, which means it explains more of the variance in Air Quality.

Classification (kNN)

4.1
Using the following 3 regions as targets: Former Soviet States, Greater Middle East, Sub-Saharan Africa, k=3 had the highest accuracy (0.6667) when the predictors were log10(population), log10(gdp) and the air quality score.

```
                      y.test
knn_3                  Former Soviet States Greater Middle East Sub-Saharan Africa
  Former Soviet States                    1                   0                  0
  Greater Middle East                     1                   1                  2
  Sub-Saharan Africa                      0                   2                  8
```

4.2
The new variable was H2O (quality of drinking water and sanitation) instead of the air quality. Everything else was kept identical. The accuracy was 0.7333.

```
                      y_2.test
knn_second             Former Soviet States Greater Middle East Sub-Saharan Africa
  Former Soviet States                    1                   0                  0
  Greater Middle East                     2                   1                  1
  Sub-Saharan Africa                      1                   0                  9
```

4.3
The second model has a higher accuracy and thus performs better. It seems that the quality of drinking water and sanitation is a better predictor than the air quality for predicting the region.